NON-WSQ COURSE SLIDES

ISO 9001:2015 Quality Management System

Course Code: C239

Tertiary Infotech Academy Pte Ltd | UEN: 201200696W

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COURSE ADMINISTRATION

Welcome and course orientation

One day | 7.5 hours | beginner level

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About the Trainer



Your Trainer

Complete this profile before class

NAME

_____

TITLE / DESIGNATION

_____

QUALIFICATIONS

_____

AREAS OF EXPERTISE

_____

TRAINING AND INDUSTRY EXPERIENCE

_____

CONTACT

_____

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer | Tertiary Infotech Academy

ROLE
Principal Trainer and course developer

EXPERTISE
Quality systems, business process improvement and applied AI

DELIVERS
Professional and technical programmes

FOUNDER
Tertiary Infotech / Tertiary Courses

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Let's know each other

1. Name, organisation and role
2. One process whose quality matters to your work
3. A recurring problem you would like to prevent
4. What success from this course would look like

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HOUSEKEEPING

Ground rules

1. **Be present**
   Return from breaks on time and stay for the full day.

2. **Participate**
   Ask, challenge and contribute evidence.

3. **Respect**
   One conversation at a time; disagree professionally.

4. **Protect information**
   Use fictional or sanitised workplace data.

5. **Stay practical**
   Link every idea to a process or decision.

6. **Own the result**
   Verify your activity output before submission.

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SCHEDULE

Lesson plan: 1 day

Morning

1. 9:30-9:50 | Administration and orientation
2. 9:50-11:45 | Foundations, Activities 1-2
3. 11:45-12:45 | Planning, support and operation
4. 12:45-1:15 | Lunch

Afternoon

1. 1:15-2:25 | Activities 3-4
2. 2:25-4:10 | Evaluation, Activities 5-6
3. 4:10-5:20 | Improvement and Activities 7-8
4. 5:20-5:30 | Recap and course feedback

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F

FOUNDATIONS

Quality management that improves real work

Purpose, principles, process approach, PDCA and current ISO 9001 status

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QUALITY MINDSET

Quality is not inspection at the end.

It is the capability of the whole system to fulfil requirements consistently and learn from evidence.

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Quality: four complementary lenses

1. **Conformity**
   Does the output meet defined requirements?

2. **Fitness for use**
   Will it work for the customer's intended purpose?

3. **Consistency**
   Can the process reproduce the intended result?

4. **Satisfaction**
   How well are customer needs and expectations fulfilled?

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Quality is not the same as grade

Quality

1. Quality asks whether requirements are fulfilled.
2. A basic product can be high quality if it reliably meets its specification.
3. Quality includes product, service, process and system characteristics.

Grade

1. Grade describes category or class with different feature sets.
2. A premium product can still be poor quality when it fails its stated requirements.
3. Do not confuse more features with better conformity.

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THREE LEVELS

Quality control, quality assurance and QMS

Quality Control

- Detect output variation
- Inspection and testing
- Product/service focus

Quality Assurance

- Build confidence in controls
- Prevention and process evidence
- Process focus

Quality Management System

- Policies, processes and roles
- Resources and evidence
- System-wide improvement

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BUSINESS VALUE

Why quality matters

1. **Customer trust**
   Consistent fulfilment reduces complaints and strengthens confidence.

2. **Cost**
   Prevention and capable processes reduce scrap, rework, delay and warranty cost.

3. **Risk**
   Controlled work lowers regulatory, safety, contractual and reputation exposure.

4. **Growth**
   Credible systems support tenders, supply-chain approval and repeat business.

5. **People**
   Clear ownership and competent work reduce firefighting.

6. **Learning**
   Evidence turns recurring problems into improvement knowledge.

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Cost of quality

Conformance cost

1. Prevention: planning, training, mistake-proofing
2. Appraisal: inspection, test, audit
3. Investment should increase capability and early detection

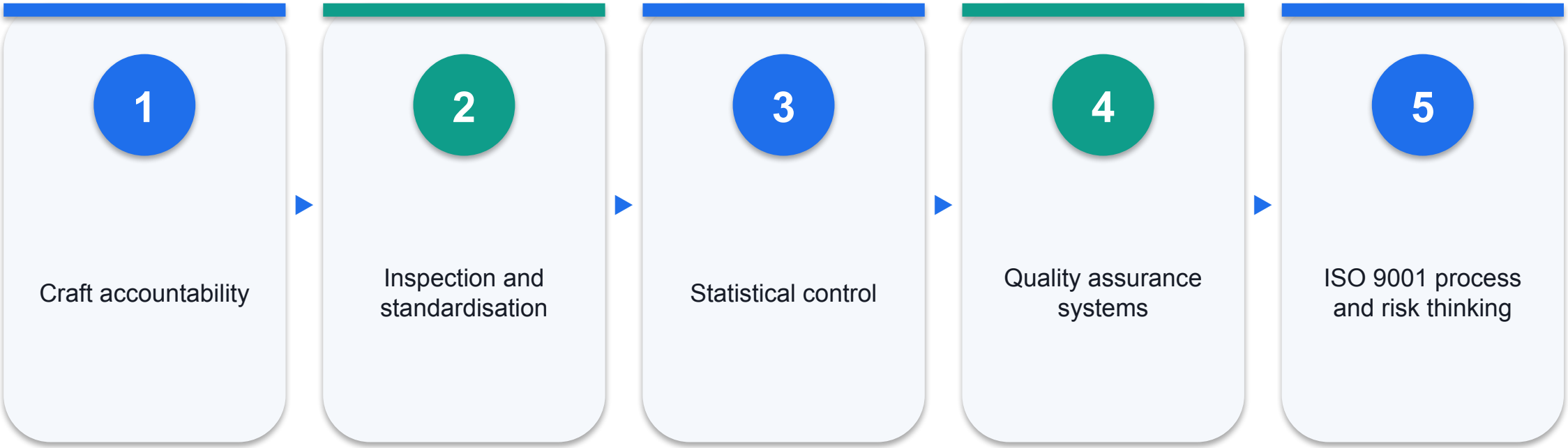
Nonconformance cost

1. Internal failure: scrap, rework, downtime
2. External failure: complaints, returns, warranty, lost trust
3. Failure cost often rises sharply after delivery

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HISTORY

How modern quality management evolved



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The ISO 9000 family

ISO 9000

- Fundamentals
- Vocabulary
- Quality management principles

ISO 9001

- QMS requirements
- Only certifiable family standard
- Applicable to any sector

Supporting guidance

- ISO 9004 sustained success
- ISO 19011 auditing guidance
- ISO/TS 9002 application guidance

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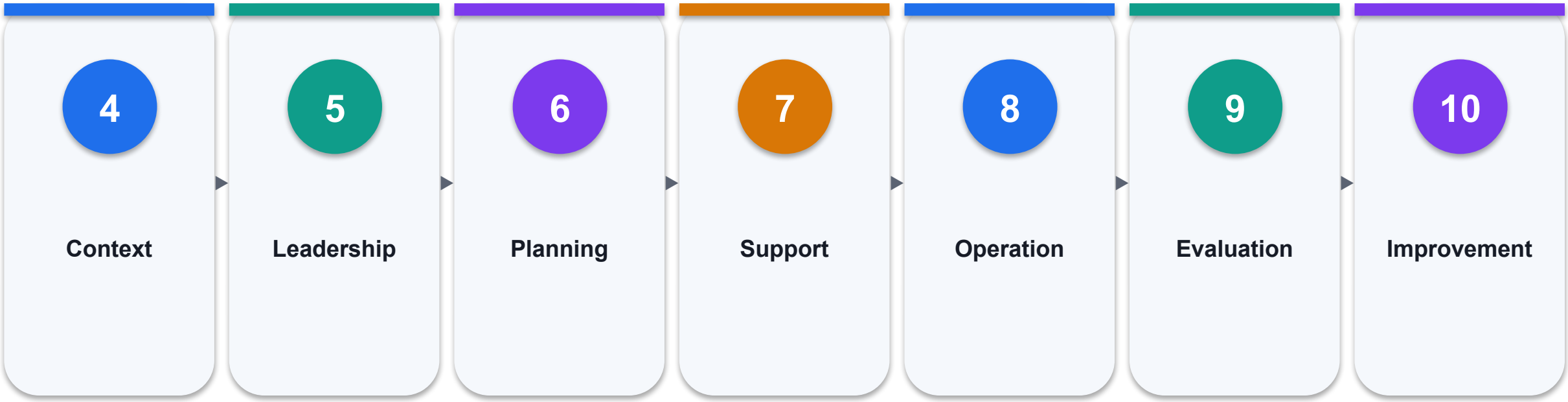
Current standard status - 12 August 2026

1. ISO 9001:2015 with Amendment 1:2024 (climate action)
2. The 2015 edition remains current as at 12 August 2026.
3. ISO states that a revised edition is expected in September 2026; teach only published requirements and plan a gap review after publication.
4. Do not treat a draft as a requirement; prepare to compare the published edition and transition guidance after release.

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CLAUSES 4-10

ISO 9001 structure: from context to improvement



Clauses 1-3 set scope, references and vocabulary; auditable QMS requirements sit in clauses 4-10.

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Seven quality management principles

1. **Customer focus**
   Understand needs, deliver value and monitor satisfaction.
2. **Leadership**
   Create unity of purpose, direction and enabling conditions.
3. **Engagement of people**
   Competent, empowered people create and deliver value.
4. **Process approach**
   Manage activities as connected processes within a coherent system.
5. **Improvement**
   Make improvement an ongoing organisational objective.
6. **Evidence-based decisions**
   Use analysis and reliable information to reduce uncertainty.
7. **Relationship management**
   Manage relevant stakeholder relationships for sustained success.

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PRINCIPLE 1

Customer focus in practice

Strong practice

1. Translate needs into clear requirements
2. Review changes before commitment
3. Monitor experience across relevant segments
4. Close the loop on complaints

Decision cautions

1. Customer says so is not the only evidence
2. Short-term satisfaction cannot override legal duties
3. Averages can hide critical segments
4. Feedback without action erodes trust

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Leadership and engagement

Enable the system

1. Connect quality direction to strategy
2. Assign authority and resources
3. Make it safe to surface problems
4. Recognise contributions to improvement

Failure patterns

1. Quality delegated to one function
2. Targets reward speed over conformity
3. Training is offered without empowerment
4. Bad news is punished or hidden

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Process, evidence and improvement

Disciplined practice

1. Define purpose, owner, inputs and outputs
2. Select measures that support decisions
3. Test changes using PDCA
4. Standardise what works

Common traps

1. Department goals conflict
2. Data definitions drift
3. Solutions start before causes are known
4. Projects close without sustainment

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PRINCIPLE 7

Relationship management

Value-creating relationships

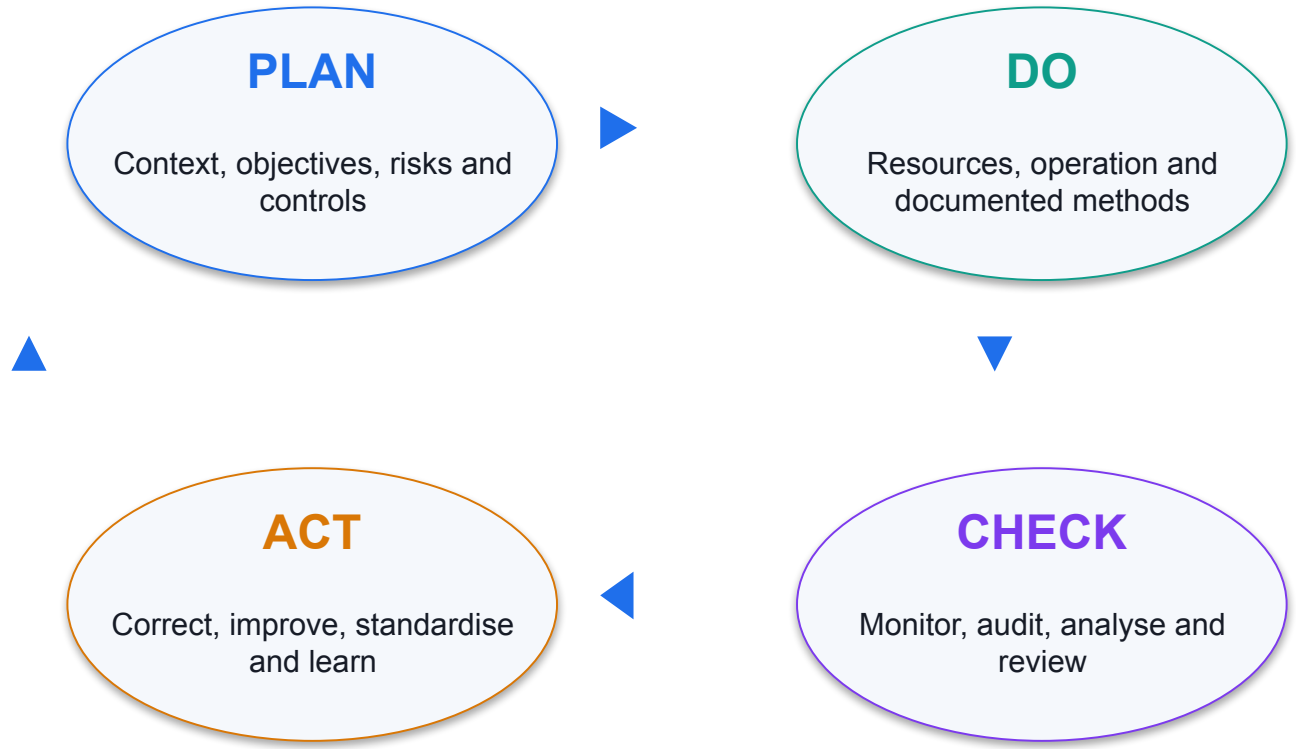
1. Segment providers by risk and importance
2. Share relevant requirements and performance
3. Develop capability where mutual value exists
4. Review relationship outcomes

Weak practice

1. Lowest price dominates all decisions
2. Supplier problems are hidden until failure
3. Requirements are ambiguous
4. Outsourcing is treated as transferring accountability

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PDCA: the operating rhythm of a QMS



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PDCA at process and system level

The cycle

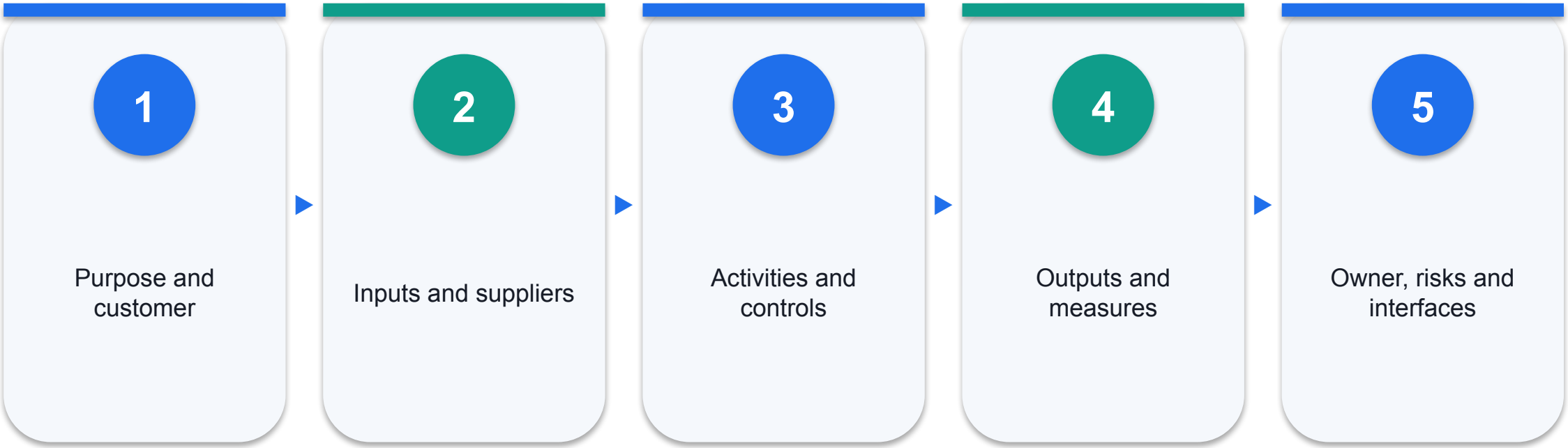
1. Plan: purpose, requirements, resources, risks, measures
2. Do: operate under controlled conditions
3. Check: compare evidence with intended results
4. Act: correct, improve and standardise

Where it applies

1. One process: order review
2. End-to-end value stream: order to delivery
3. Whole QMS: clauses 4-10
4. Improvement portfolio: priorities and benefits

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Process approach: define before you improve



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Certification: what it is and is not

Voluntary framework

- Certification is not required by ISO 9001
- Customers or tenders may require it
- Organisations may use the QMS without certification

Independent assessment

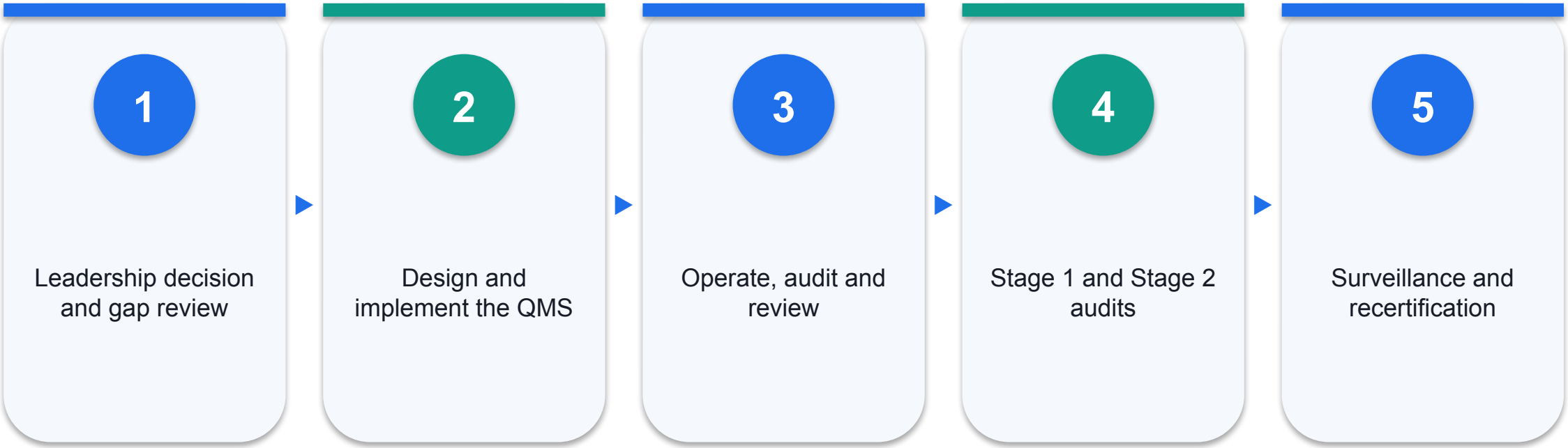
- ISO does not certify organisations
- Accredited certification bodies audit
- Scope and conformity must be demonstrated

Ongoing cycle

- Stage 1 readiness review
- Stage 2 implementation audit
- Surveillance and recertification

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Typical certification journey



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ANNEX SL

Integrated management systems

Integration advantages

1. Common harmonised structure
2. Shared context and interested parties
3. Common governance, audit and review
4. Reduced duplication across systems

Examples

1. ISO 9001 | quality
2. ISO 14001 | environment
3. ISO 45001 | occupational health and safety
4. ISO/IEC 27001 | information security

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1

PART 1

Context, leadership, planning, support and operation

Quality foundations, context, leadership, planning, support and operation

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CLAUSE 4.1

4.1 Understanding the organisation and its context: intent and control

1

Intent

Determine internal and external issues that affect QMS outcomes.

2

Control 1

Review strategic, market, technology, regulatory and workforce issues.

3

Control 2

Decide whether climate change is relevant under Amendment 1:2024.

4

Control 3

Refresh the analysis when conditions change.

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CLAUSE 4.1

4.1 Understanding the organisation and its context: evidence and judgement

Useful objective evidence

1. Strategy or risk review
2. Context register with owners
3. Trend and regulatory scan
4. Management review discussion

Common weak practice

1. Static SWOT never revisited
2. Generic issues unrelated to QMS results
3. Climate relevance asserted without rationale
4. No link from issue to action

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CLAUSE 4.2

4.2 Interested parties and requirements: intent and control

1

Intent

Identify relevant interested parties and requirements that affect the QMS.

2

Control 1

Select parties because they affect consistent conformity or satisfaction.

3

Control 2

Capture applicable customer, statutory and regulatory needs.

4

Control 3

Monitor changes, including climate-related expectations where relevant.

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CLAUSE 4.2

4.2 Interested parties and requirements: evidence and judgement

Useful objective evidence

1. Customer contracts
2. Regulatory obligations register
3. Supplier or employee requirements
4. Review records showing updates

Common weak practice

1. Listing everyone without relevance
2. Confusing wishes with binding requirements
3. No owner for monitoring changes
4. Requirements not translated into processes

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CLAUSE 4.3

4.3 Scope of the QMS: intent and control

1

Intent

Set a clear, truthful boundary for products, services, sites and processes.

2

Control 1

Consider context and interested-party requirements.

3

Control 2

State products, services and organisational boundaries.

4

Control 3

Justify any requirement considered not applicable.

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CLAUSE 4.3

4.3 Scope of the QMS: evidence and judgement

Useful objective evidence

1. Approved scope statement
2. Site and process boundary map
3. Applicability rationale
4. Certification scope aligned with operations

Common weak practice

1. Marketing scope broader than reality
2. Exclusions used to avoid responsibility
3. Outsourced processes treated as out of scope
4. Scope not available as controlled information

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CLAUSE 4.4

4.4 QMS and its processes: intent and control

1

Intent

Manage interconnected processes so they achieve intended results.

2

Control 1

Define inputs, outputs, sequence and interaction.

3

Control 2

Assign criteria, measures, resources, owners and risks.

4

Control 3

Evaluate and improve process performance.

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4.4 QMS and its processes: evidence and judgement

Useful objective evidence

1. SIPOC and process maps
2. Process KPIs and targets
3. Owner and responsibility records
4. Records of review and improvement

Common weak practice

1. Procedure library with no process model
2. Local optimisation damages end-to-end flow
3. Activity measures replace outcome measures
4. No evidence that planned results are achieved

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CLAUSE 5.1

5.1 Leadership and commitment: intent and control

1

Intent

Top management owns QMS effectiveness and integrates it into business direction.

2

Control 1

Align quality with strategy and business processes.

3

Control 2

Provide resources and support process owners.

4

Control 3

Promote customer focus, process approach and improvement.

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CLAUSE 5.1

5.1 Leadership and commitment: evidence and judgement

Useful objective evidence

1. Decisions and resource allocation
2. Leadership participation in reviews
3. Quality objectives linked to strategy
4. Evidence of removing barriers

Common weak practice

1. Delegating the QMS to the quality manager
2. Talking quality but rewarding speed only
3. Reviews without decisions
4. Customer risk absent from leadership agenda

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CLAUSE 5.2

5.2 Quality policy: intent and control

1

Intent

Provide a direction that fits the organisation and supports quality objectives.

2

Control 1

Make commitments to applicable requirements and improvement.

3

Control 2

Communicate, understand and apply the policy.

4

Control 3

Keep it available as controlled information.

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5.2 Quality policy: evidence and judgement

Useful objective evidence

1. Policy linked to business purpose
2. Interview evidence of meaning
3. Objectives derived from commitments
4. Current approved copy

Common weak practice

1. Generic poster language
2. Employees can recite but not apply it
3. Policy disconnected from objectives
4. Obsolete versions remain in use

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CLAUSE 5.3

5.3 Roles, responsibilities and authorities: intent and control

1

Intent

Make accountability clear so processes deliver and QMS results are reported.

2

Control 1

Assign process ownership and decision authority.

3

Control 2

Ensure reporting on performance and improvement.

4

Control 3

Preserve customer focus during organisational change.

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CLAUSE 5.3

5.3 Roles, responsibilities and authorities: evidence and judgement

Useful objective evidence

1. RACI or role profiles
2. Process-owner terms of reference
3. Escalation paths
4. Management reporting records

Common weak practice

1. Everyone is responsible means nobody owns it
2. Responsibility without authority
3. Quality team owns operational controls
4. Gaps during leave or restructuring

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Diagnose Quality Gaps from Evidence

ACTIVITY 1

Analyse LPC's complaint, defect and delivery evidence; prioritise improvement areas and connect them to customer impact.

EXPECTED OUTPUT

A completed quality-gap prioritisation matrix and evidence-backed problem statement.

Resources: Scenario brief PDF, Quality Gap Workbook (Excel), team discussion

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Diagnose Quality Gaps from Evidence

?

LPC's customer complaints doubled in six months. Late deliveries, burr defects and inconsistent certificates are repeatedly mentioned, but managers disagree about the main cause.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Diagnose Quality Gaps from Evidence

CHECK YOUR EVIDENCE

The top three gaps are traceable to supplied evidence; the problem statement contains a baseline, affected process and measurable effect.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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Map and Improve the Order-to-Delivery Process

ACTIVITY 2

Map LPC's order-to-delivery process, expose hand-off risks and design a controlled future state.

EXPECTED OUTPUT

A SIPOC, current-state map, improvement hypothesis and PDCA action plan.

Resources: Process Mapping Workbook (Excel), SIPOC checklist PDF

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ACTIVITY 2 | DECISION PROMPT

Map and Improve the Order-to-Delivery Process



Sales accepts emailed drawings, Planning creates job travellers, Production machines parts, Quality inspects and Logistics ships. Rework and missing revision information frequently surface late.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Map and Improve the Order-to-Delivery Process

CHECK YOUR EVIDENCE

The map has clear boundaries, owners, inputs, outputs, controls and measures; the PDCA experiment has a testable prediction.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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2

TOPIC 2

Planning, support and operation

Risk-based planning | resources | documented information | operational control

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CLAUSE 6.1

6.1 Actions for risks and opportunities: intent and control

1

Intent

Plan proportionate actions that protect intended results and enable improvement.

2

Control 1

Identify risks and opportunities from context and processes.

3

Control 2

Integrate actions into operational controls.

4

Control 3

Evaluate whether actions work.

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CLAUSE 6.1

6.1 Actions for risks and opportunities: evidence and judgement

Useful objective evidence

1. Risk and opportunity register
2. Changed controls or plans
3. Owners and completion evidence
4. Effectiveness review

Common weak practice

1. Risk register isolated from work
2. Every risk scored but none acted upon
3. Controls copied from templates
4. Completion confused with effectiveness

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CLAUSE 6.2

6.2 Quality objectives and plans: intent and control

1

Intent

Translate policy into measurable outcomes with a credible achievement plan.

2

Control 1

Set relevant measurable objectives at appropriate levels.

3

Control 2

Define actions, resources, owner, timing and evaluation method.

4

Control 3

Update objectives when circumstances change.

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6.2 Quality objectives and plans: evidence and judgement

Useful objective evidence

1. Objective and KPI dictionary
2. Baseline and target rationale
3. Action plans with owners
4. Periodic results and decisions

Common weak practice

1. Vague goals such as improve quality
2. Targets without baselines
3. KPIs without decisions
4. Measures that encourage gaming

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6.3 Planning changes: intent and control

1

Intent

Change the QMS deliberately without losing integrity or control.

2

Control 1

Define purpose and likely consequences.

3

Control 2

Check resources and responsibilities.

4

Control 3

Plan sequencing, validation and communication.

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CLAUSE 6.3

6.3 Planning changes: evidence and judgement

Useful objective evidence

1. Change request and impact analysis
2. Implementation and rollback plan
3. Competence or communication evidence
4. Post-change effectiveness review

Common weak practice

1. Emergency change becomes permanent
2. No risk assessment
3. Documents updated after rollout
4. Interfaces and dependent processes missed

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CLAUSE 7.1

7.1 Resources: intent and control

1

Intent

Provide and maintain people, infrastructure, environment, measurement resources and knowledge.

2

Control 1

Determine resource needs using process and performance evidence.

3

Control 2

Protect measurement validity and traceability where needed.

4

Control 3

Maintain organisational knowledge through change.

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CLAUSE 7.1

7.1 Resources: evidence and judgement

Useful objective evidence

1. Capacity and maintenance plans
2. Calibration or verification records
3. Work-environment controls
4. Knowledge capture and succession plans

Common weak practice

1. Resources approved without need analysis
2. Calibration status unknown
3. Environment affects conformity
4. Critical know-how held by one person

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CLAUSE 7.2-7.4

7.2-7.4 Competence, awareness and communication: intent and control

1

Intent

Ensure people can perform work, understand impact and receive the right information.

2

Control 1

Define competence by role and risk.

3

Control 2

Evaluate training or other actions for effectiveness.

4

Control 3

Plan what, when, with whom and how to communicate.

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CLAUSE 7.2-7.4

7.2-7.4 Competence, awareness and communication: evidence and judgement

Useful objective evidence

1. Competence matrix and criteria
2. Observed performance after training
3. Awareness interviews
4. Communication plan and records

Common weak practice

1. Attendance certificate treated as competence
2. No effectiveness check
3. Broadcast email treated as communication
4. Contractors left outside awareness controls

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CLAUSE 7.5

7.5 Documented information: intent and control

1. **Intent**
   Create and control information needed to operate processes and prove results.

2. **Control 1**
   Identify, describe, review and approve information appropriately.

3. **Control 2**
   Make current information available where needed.

4. **Control 3**
   Protect, retain, retrieve and dispose of records.

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7.5 Documented information: evidence and judgement

Useful objective evidence

1. Document register and approvals
2. Point-of-use revision controls
3. Access and backup evidence
4. Retention and disposal records

Common weak practice

1. Documentation for its own sake
2. Obsolete copies at point of use
3. Records editable without traceability
4. External documents uncontrolled

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CLAUSE 8.1

8.1 Operational planning and control: intent and control

1

Intent

Plan and control the work required to meet product and service requirements.

2

Control 1

Set process criteria and acceptance criteria.

3

Control 2

Provide resources and controlled information.

4

Control 3

Control planned changes and review unintended changes.

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CLAUSE 8.1

8.1 Operational planning and control: evidence and judgement

Useful objective evidence

1. Control plan or service plan
2. Work instructions and acceptance criteria
3. Production or service records
4. Change and deviation approvals

Common weak practice

1. Plans exist but are not followed
2. Acceptance criteria subjective
3. Outsourced work omitted
4. Unplanned changes not reviewed

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CLAUSE 8.2

8.2 Requirements for products and services: intent and control

1

Intent

Understand, review and communicate requirements before committing to supply.

2

Control 1

Control customer communication.

3

Control 2

Determine stated, unstated, legal and organisational requirements.

4

Control 3

Review feasibility and control requirement changes.

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CLAUSE 8.2

8.2 Requirements for products and services: evidence and judgement

Useful objective evidence

1. Contract or order review
2. Customer communication records
3. Technical feasibility review
4. Revision and change acknowledgement

Common weak practice

1. Sales commits before review
2. Assumptions remain undocumented
3. Changed requirements not cascaded
4. Complaint feedback not used

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CLAUSE 8.3

8.3 Design and development: intent and control

1

Intent

Control design stages, inputs, reviews, verification, validation, outputs and changes where applicable.

2

Control 1

Plan stages and responsibilities proportionately.

3

Control 2

Make inputs complete and unambiguous.

4

Control 3

Use reviews, verification and validation for distinct purposes.

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CLAUSE 8.3

8.3 Design and development: evidence and judgement

Useful objective evidence

1. Design plan and input record
2. Review and verification evidence
3. Validation against intended use
4. Approved outputs and change history

Common weak practice

1. Verification confused with validation
2. Design changes bypass control
3. Customer-use conditions missed
4. Outputs inadequate for later processes

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CLAUSE 8.4

8.4 Externally provided processes, products and services: intent and control

1

Intent

Control suppliers and outsourced work according to their effect on conformity.

2

Control 1

Select, evaluate, monitor and re-evaluate providers.

3

Control 2

Set type and extent of control using risk and performance.

4

Control 3

Communicate clear requirements to providers.

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CLAUSE 8.4

8.4 Externally provided processes, products and services: evidence and judgement

Useful objective evidence

1. Approved-provider criteria
2. Supplier performance trends
3. Purchase specifications
4. Incoming verification and re-evaluation records

Common weak practice

1. Approval based only on price
2. Same controls for all suppliers
3. Outsourcing treated as transferring accountability
4. Poor performance not acted upon

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CLAUSE 8.5

8.5 Production and service provision: intent and control

1

Intent

Deliver under controlled conditions while protecting identity, traceability, property and preservation.

2

Control 1

Provide current information and competent people.

3

Control 2

Use suitable monitoring, identification and release controls.

4

Control 3

Control changes and post-delivery needs.

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CLAUSE 8.5

8.5 Production and service provision: evidence and judgement

Useful objective evidence

1. Job traveller or service record
2. Identification and traceability
3. Customer-property log
4. Change authorisation and preservation evidence

Common weak practice

1. Tribal knowledge replaces controls
2. Traceability breaks at rework
3. Customer data or property mishandled
4. Temporary process change undocumented

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8.6 Release of products and services: intent and control

1

Intent

Release only after planned acceptance arrangements are satisfactorily completed.

2

Control 1

Verify acceptance evidence before release.

3

Control 2

Retain evidence of conformity.

4

Control 3

Identify the person authorising release.

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8.6 Release of products and services: evidence and judgement

Useful objective evidence

1. Inspection or service acceptance record
2. Test results against criteria
3. Release status
4. Authorised release signature or system record

Common weak practice

1. Schedule pressure overrides acceptance
2. Evidence completed after shipment
3. Authority unclear
4. Concession not approved

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CLAUSE 8.7

8.7 Control of nonconforming outputs: intent and control

1

Intent

Prevent unintended use or delivery and control what happens next.

2

Control 1

Identify and segregate or otherwise control the output.

3

Control 2

Choose correction, containment, concession or other disposition.

4

Control 3

Re-verify after correction and retain decision evidence.

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CLAUSE 8.7

8.7 Control of nonconforming outputs: evidence and judgement

Useful objective evidence

1. Nonconformance record
2. Identification and segregation
3. Disposition and concession approval
4. Re-verification evidence

Common weak practice

1. Rework performed without approval
2. Correction hides systemic cause
3. Affected scope not traced
4. Customer notified too late

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Risk-based thinking: prioritise proportionately

IMPACT

5	10	15	20	25
4	8	12	16	20
3	6	9	12	15
2	4	6	8	10
1	2	3	4	5

LIKELIHOOD

Determine What could affect the intended result?

Plan Actions proportionate to potential impact

Integrate Embed actions in real processes

Evaluate Check whether actions are effective

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CLAUSE 6.2

Quality objective design test

SMART and owned

1. Specific outcome and population
2. Measurable definition and baseline
3. Target and due date
4. Owner, resources and action plan

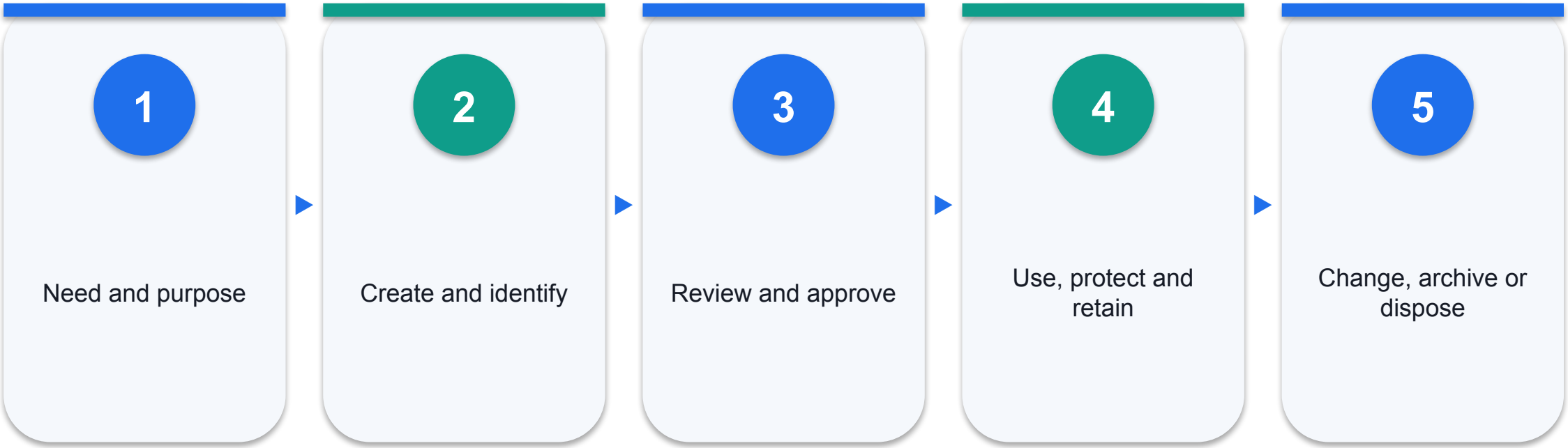
Decision-ready

1. Relevant to policy and process result
2. Balanced leading and lagging measures
3. Review cadence and response threshold
4. Evidence of change and effectiveness

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CLAUSE 7.5

Documented information life cycle



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Selecting the right quality tool

Understand

- Flowchart or SIPOC
- Check sheet
- Stratification

Prioritise and explain

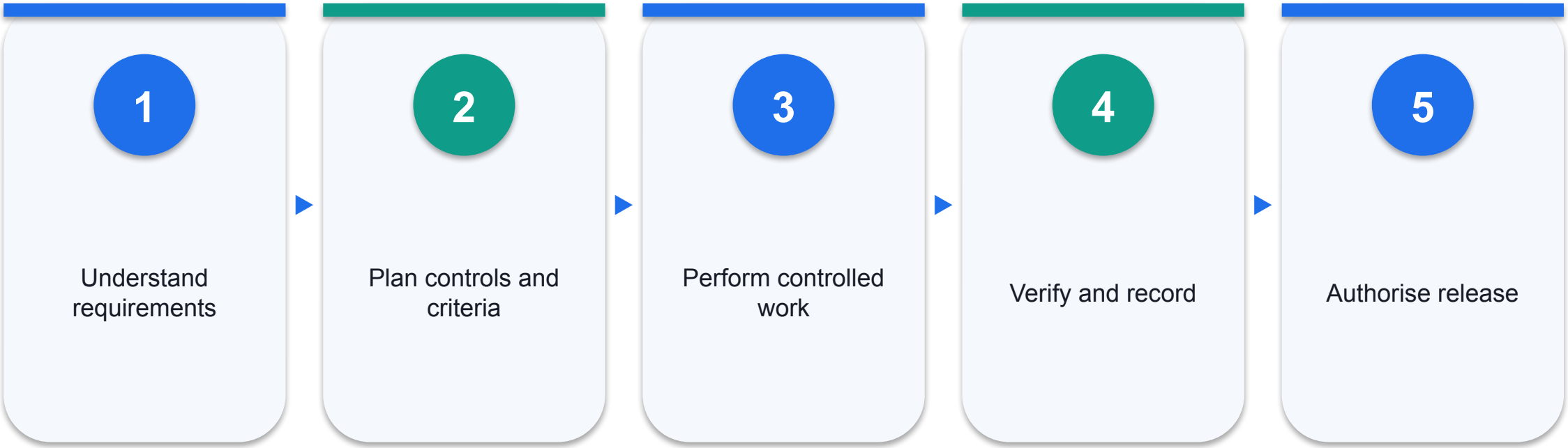
- Pareto analysis
- Cause-and-effect
- 5 Whys

Control and improve

- Control chart
- FMEA
- PDCA or A3 report

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Operational control: requirement to release



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SECTOR APPLICATION

Service and manufacturing evidence

Manufacturing

1. Order review and job traveller
2. Material identification and traceability
3. Inspection, test and release record
4. Nonconforming-product control

Services

1. Service brief and acceptance criteria
2. Competence and service checklist
3. Customer communication record
4. Service recovery and feedback record

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Select and Apply Quality Tools

ACTIVITY 3

Use check sheets, Pareto analysis and cause-and-effect analysis to prioritise LPC's customer complaints.

EXPECTED OUTPUT

A coded complaint check sheet, Pareto chart and evidence-led cause hypothesis.

Resources: Quality Tools Workbook (Excel), Tool Selection Checklist PDF

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Select and Apply Quality Tools

?

LPC has 60 recent complaints from three customer segments. The descriptions are inconsistent and teams jump directly to favourite solutions.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Select and Apply Quality Tools

CHECK YOUR EVIDENCE

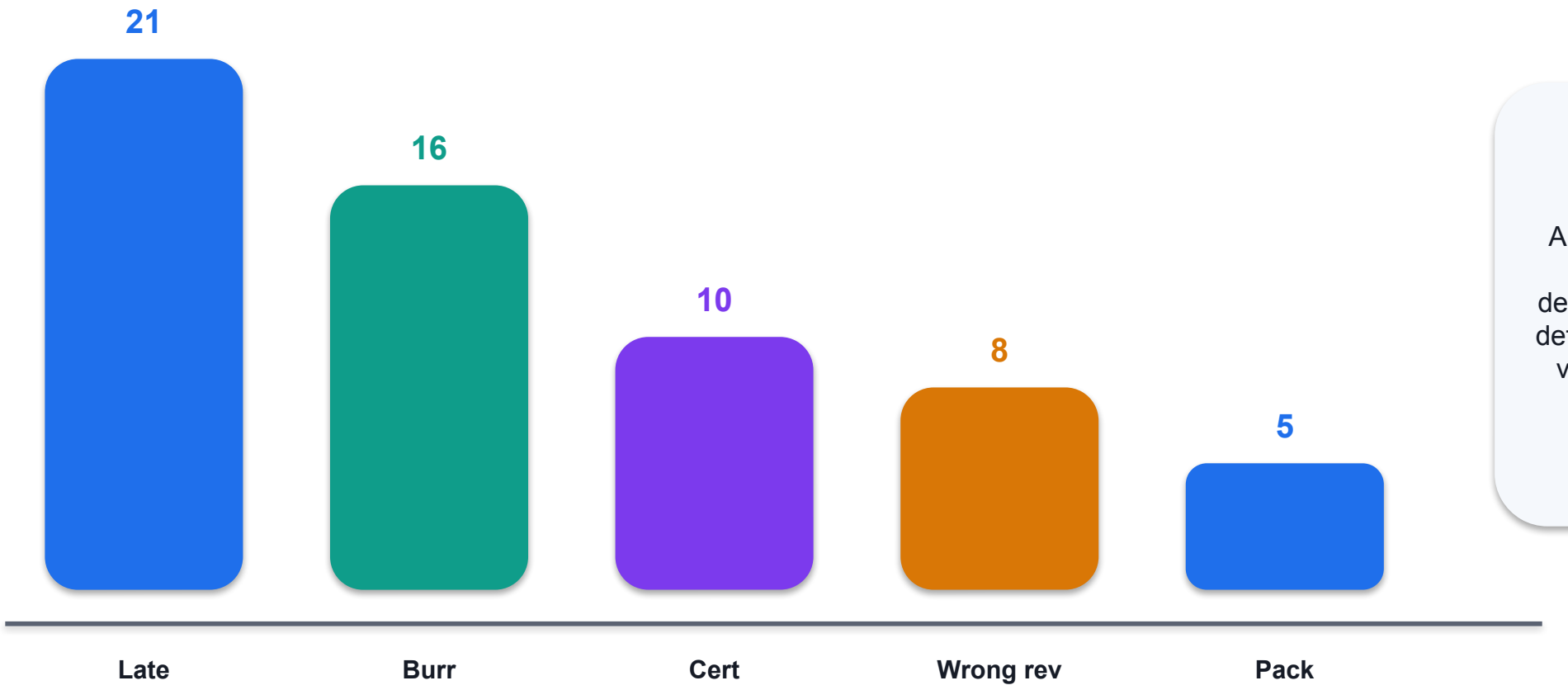
The Pareto chart reconciles to the complaint total and the selected cause hypothesis is explicitly labelled for verification.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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ACTIVITY 3 DATA

LPC complaint categories before prioritisation



READ IT

A chart highlights patterns. A decision still needs definitions, context, valid data and a response threshold.

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Set Quality Objectives and KPIs

ACTIVITY 4

Translate LPC's quality policy into SMART objectives, operational measures and a review cadence.

EXPECTED OUTPUT

A KPI dictionary and dynamic dashboard with targets, formulas, status signals and ownership.

Resources: KPI Tracker Workbook (Excel), KPI Design Checklist PDF

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ACTIVITY 4 | DECISION PROMPT

Set Quality Objectives and KPIs

?

LPC's current dashboard lists activity counts but no targets, owners, decision thresholds or links to customer outcomes.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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ACTIVITY 4 | VERIFY

Set Quality Objectives and KPIs

CHECK YOUR EVIDENCE

Every KPI has an auditable formula, source, owner, target and defined management response; workbook formulas return no errors.

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CHECK YOUR UNDERSTANDING

Part 1 recap

1. **Context**
   Know the issues, parties, scope and processes.

2. **Leadership**
   Direction, accountability and customer focus.

3. **Planning**
   Risks, objectives and controlled change.

4. **Support**
   Resources, competence, communication and information.

5. **Operation**
   Controlled work from requirement through release.

6. **Evidence**
   Use information that supports decisions and traceability.

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2

PART 2

Performance evaluation and continual improvement

Performance evaluation, audit, management review and continual improvement

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TOPIC 3

Performance evaluation

Monitoring | analysis | internal audit | management review

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CLAUSE 9.1

9.1 Monitoring, measurement, analysis and evaluation: intent and control

1

Intent

Decide what to measure and use valid data to evaluate QMS performance.

2

Control 1

Define method, timing and criteria for analysis.

3

Control 2

Evaluate customer satisfaction and process results.

4

Control 3

Use evidence to judge effectiveness and improvement needs.

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CLAUSE 9.1

9.1 Monitoring, measurement, analysis and evaluation: evidence and judgement

Useful objective evidence

1. KPI definitions and raw data
2. Customer feedback trends
3. Analysis with decisions
4. Calibration or data-quality evidence

Common weak practice

1. Dashboard without decisions
2. Averages hide segments
3. Activity counts replace outcomes
4. Unvalidated data drives action

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CLAUSE 9.2

9.2 Internal audit: intent and control

1

Intent

Provide objective information on conformity and effective implementation.

2

Control 1

Plan a risk-based audit programme.

3

Control 2

Define scope and criteria for each audit.

4

Control 3

Use impartial auditors, report results and follow up actions.

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CLAUSE 9.2

9.2 Internal audit: evidence and judgement

Useful objective evidence

1. Audit programme and plan
2. Process-based checklist
3. Evidence notes and findings
4. Correction and corrective-action follow-up

Common weak practice

1. Auditing documents only
2. Auditor checks own work
3. Checklist treated as evidence
4. Findings prescribe consultant solutions

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CLAUSE 9.3

9.3 Management review: intent and control

1

Intent

Top management evaluates continuing suitability, adequacy, effectiveness and strategic alignment.

2

Control 1

Review required inputs as a connected performance story.

3

Control 2

Decide improvement, changes and resource needs.

4

Control 3

Retain outputs, owners and follow-up evidence.

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CLAUSE 9.3

9.3 Management review: evidence and judgement

Useful objective evidence

1. Agenda and input pack
2. Minutes with decisions
3. Resource and change actions
4. Closure tracking

Common weak practice

1. Data presentation with no decisions
2. Missing trend and context
3. Actions without owners
4. Review delegated away from top management

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CLAUSE 9.1

A useful measure has six parts

1. **Purpose**
   Which decision will it support?
2. **Definition**
   What exactly is counted and excluded?
3. **Method**
   How is it collected and calculated?
4. **Owner**
   Who validates and responds?
5. **Threshold**
   When is intervention required?
6. **Cadence**
   How often is it reviewed?

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Leading and lagging indicators

Leading signals

1. Training effectiveness observation
2. Preventive-maintenance adherence
3. Supplier-risk action closure
4. First-pass review completion

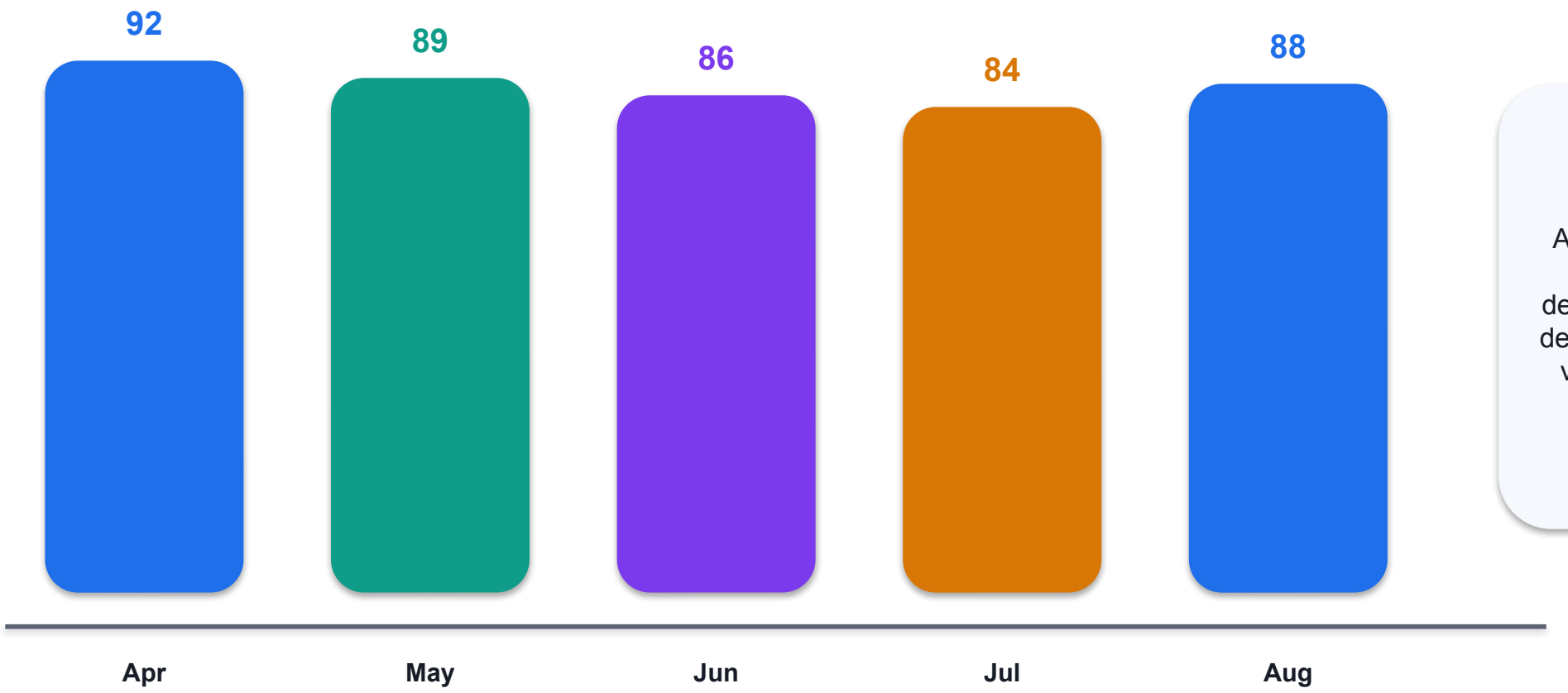
Lagging results

1. Complaint rate
2. Defect or rework rate
3. On-time delivery
4. Warranty or service recovery cost

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EVALUATE THE TREND

Monthly on-time delivery trend



READ IT

A chart highlights patterns. A decision still needs definitions, context, valid data and a response threshold.

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CLAUSE 9.1.2

Customer satisfaction evidence

Evidence sources

1. Structured surveys
2. Complaint and return trends
3. Repeat business and retention
4. Delivery and service outcomes

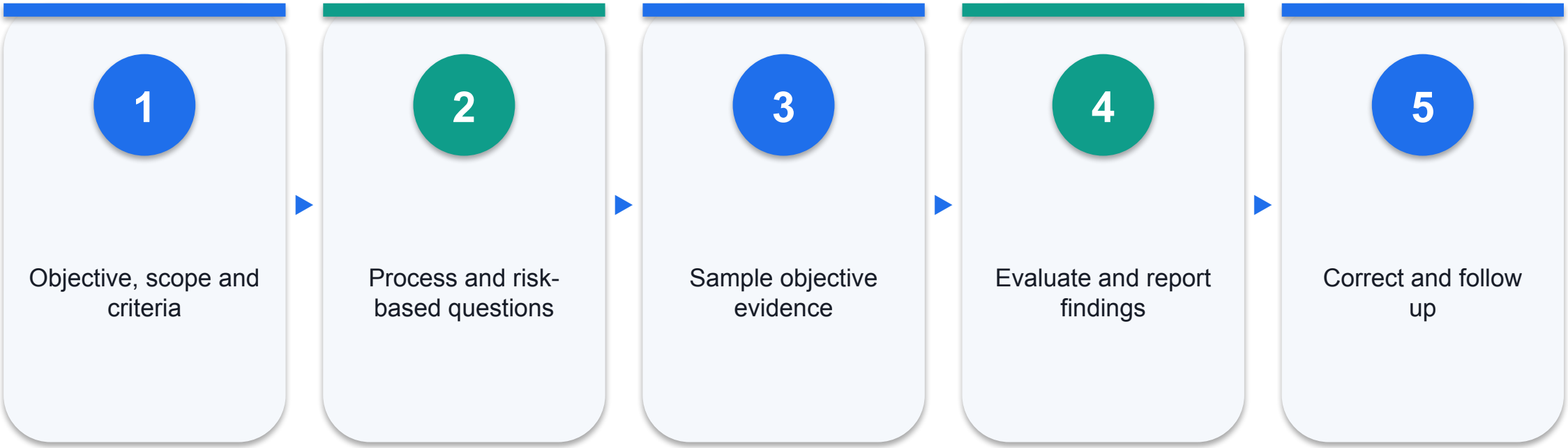
Interpretation risks

1. Response bias
2. Low or changing sample size
3. Averages hiding segments
4. Metrics without qualitative context

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CLAUSE 9.2

Internal audit trail



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Audit evidence triangulation

Documents

- What should happen
- Current approved method
- Defined criteria

Interviews

- What people understand
- How decisions are made
- Where variation occurs

Observation and records

- What actually happened
- Traceable samples
- Results and effectiveness

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Write a defensible nonconformity

Required structure

1. Requirement: the applicable criterion
2. Evidence: specific, traceable facts
3. Gap: how evidence fails the criterion
4. Scope: affected sample or process

Avoid

1. Vague: documentation is poor
2. Opinion: staff seemed careless
3. Prescription: buy this software
4. Overreach: one sample proves total failure

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Evaluate a Process Against QMS Requirements

ACTIVITY 5

Assess LPC's supplier-control process using a requirements-to-evidence matrix without treating a checklist as proof.

EXPECTED OUTPUT

A clause applicability and evidence matrix with conformity judgements and follow-up questions.

Resources: Evaluation Matrix Workbook (Excel), Evidence Quality Checklist PDF

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Evaluate a Process Against QMS Requirements



Supplier approvals exist, but several files are incomplete and incoming defects have increased. Procurement says the process is compliant because a procedure exists.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Evaluate a Process Against QMS Requirements

CHECK YOUR EVIDENCE

Each judgement cites a requirement, evidence and result; unknowns remain evidence gaps rather than automatic nonconformities.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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Conduct an Internal Audit Simulation

ACTIVITY 6

Plan and conduct a simulated internal audit of LPC's order-review and production-control process.

EXPECTED OUTPUT

An audit plan, process-based checklist, evidence log and correctly worded findings.

Resources: Internal Audit Workbook (Excel), Audit Checklist PDF, role-play evidence pack

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Conduct an Internal Audit Simulation



A customer returned parts made to an obsolete drawing revision. LPC needs an impartial internal audit before management review.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Conduct an Internal Audit Simulation

CHECK YOUR EVIDENCE

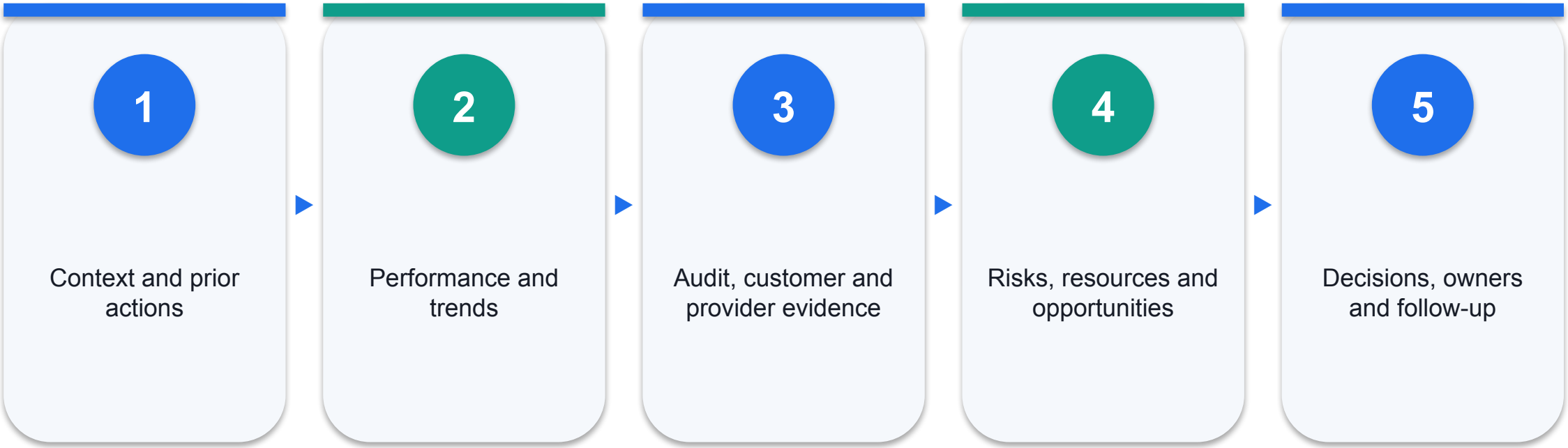
The audit trail is reproducible and every nonconformity statement includes the requirement, objective evidence and unmet condition.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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CLAUSE 9.3

Management review converts evidence into decisions



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CLAUSE 9.3

Management review inputs and outputs

Inputs

1. Changes in context and interested parties
2. Objectives, process and conformity results
3. Customer, audit and supplier performance
4. Resources, risks and improvement opportunities

Outputs

1. Improvement decisions
2. Need for QMS changes
3. Resource decisions
4. Actions, owners and completion dates

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TOPIC 4

Continual improvement

Nonconformity | root cause | corrective action | reporting

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CLAUSE 10.1

10.1 General improvement: intent and control

1

Intent

Select and act on opportunities that improve products, services, results and customer satisfaction.

2

Control 1

Use performance, audit, review and feedback evidence.

3

Control 2

Correct, prevent or reduce undesired effects.

4

Control 3

Improve QMS suitability, adequacy and effectiveness.

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10.1 General improvement: evidence and judgement

Useful objective evidence

1. Prioritised improvement pipeline
2. Business case and expected result
3. Implementation evidence
4. Benefits and lessons review

Common weak practice

1. Too many unprioritised ideas
2. Improvement equals only corrective action
3. No baseline
4. Benefits claimed without evidence

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CLAUSE 10.2

10.2 Nonconformity and corrective action: intent and control

1

Intent

React, control, eliminate causes and verify that recurrence is prevented.

2

Control 1

Correct and deal with consequences.

3

Control 2

Determine cause and whether similar issues exist.

4

Control 3

Implement action, review effectiveness and update the QMS where needed.

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CLAUSE 10.2

10.2 Nonconformity and corrective action: evidence and judgement

Useful objective evidence

1. NCR and containment record
2. Root-cause evidence
3. Action plan tied to causes
4. Effectiveness and systemic update evidence

Common weak practice

1. Correction confused with corrective action
2. Root cause is person did not follow
3. Action closes on due date alone
4. Similar risks not considered

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CLAUSE 10.3

10.3 Continual improvement: intent and control

1

Intent

Use learning and evidence to keep improving QMS suitability, adequacy and effectiveness.

2

Control 1

Use analysis and management review outputs.

3

Control 2

Standardise successful changes.

4

Control 3

Maintain learning, ownership and follow-up.

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10.3 Continual improvement: evidence and judgement

Useful objective evidence

1. Improvement portfolio
2. Before-and-after evidence
3. Updated standard work
4. Lessons learned and next review

Common weak practice

1. One-off project with no sustainment
2. Local gain harms another process
3. No control-plan update
4. No follow-up after handover

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Three different responses

Correction

- Fix the detected problem
- Rework, replace or restore
- Immediate output focus

Containment

- Control affected scope
- Protect customer and process
- Prevent unintended use

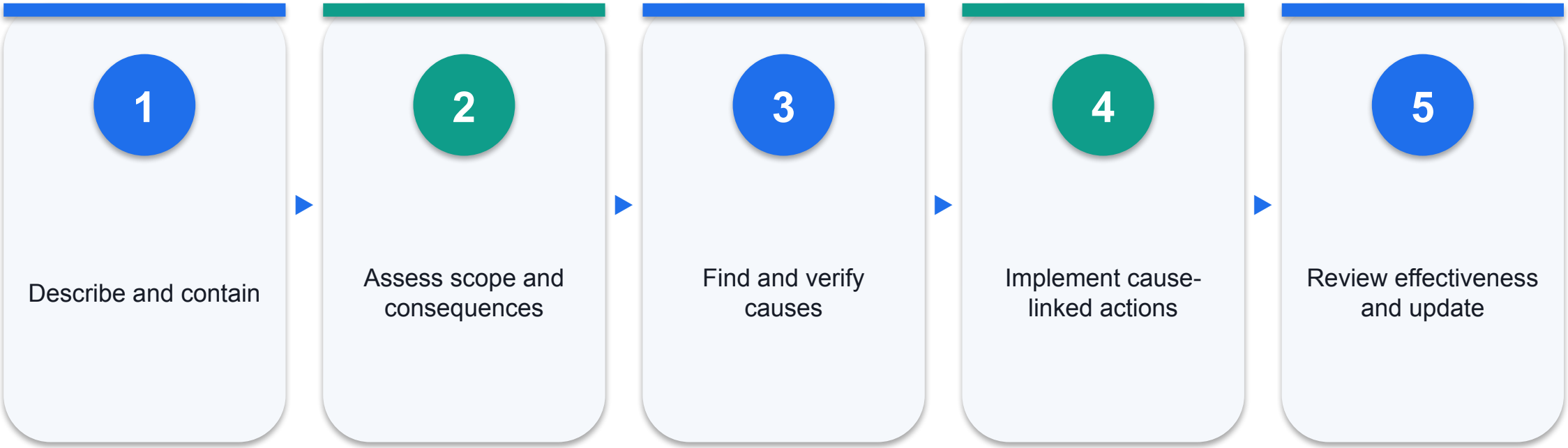
Corrective action

- Eliminate verified cause
- Prevent recurrence elsewhere
- System focus

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CLAUSE 10.2

CAPA logic



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Root cause: evidence before certainty

Useful methods

1. 5 Whys for causal chain
2. Fishbone for categories
3. Process map for hand-offs
4. Data stratification for patterns

Reasoning traps

1. Stop at human error
2. Choose favourite cause
3. Confuse correlation with cause
4. Skip verification against evidence

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Corrective-action quality test

1. **Cause linked**
   Each action addresses a verified cause.
2. **Owned**
   One accountable owner has authority and resources.
3. **Timed**
   Due date reflects risk and implementation reality.
4. **Systemic**
   Similar processes and risks are considered.
5. **Measured**
   Effectiveness criteria are defined before closure.
6. **Sustained**
   Documents, competence and controls are updated.

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VERIFY THE RESULT

Effectiveness is not completion

Implementation evidence

1. Action implemented as planned
2. Evidence exists that control is operating
3. People are competent in changed work
4. Required documents and systems updated

Effectiveness evidence

1. The problem did not recur over a meaningful period
2. Trend moved toward target
3. Audit sample confirms sustained control
4. No adverse effect transferred elsewhere

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Follow Up a Nonconformity and Corrective Action

ACTIVITY 7

Manage LPC's obsolete-drawing nonconformity using containment, 5 Whys, action planning and effectiveness criteria.

EXPECTED OUTPUT

A complete NCR/CAPA record with evidence-based root cause and an effectiveness review plan.

Resources: CAPA Workbook (Excel), Corrective Action Checklist PDF

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Follow Up a Nonconformity and Corrective Action



Twenty-four parts were produced to revision B after the customer approved revision C. Eight were shipped before the error was detected.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Follow Up a Nonconformity and Corrective Action

CHECK YOUR EVIDENCE

Actions address verified causes, not only symptoms; the effectiveness criterion is measurable and scheduled after sufficient evidence can accumulate.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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Document the Continual Improvement Report

ACTIVITY 8

Synthesize evidence from all prior activities into an accountable improvement report and management-review recommendation.

EXPECTED OUTPUT

A structured improvement report with baseline, analysis, decisions, responsibilities, evidence and lessons learned.

Resources: Improvement Report Workbook (Excel), Report Quality Checklist PDF

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Document the Continual Improvement Report



Three months after LPC launched its improvement programme, management needs evidence of results, unresolved risks and recommended next actions.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Document the Continual Improvement Report

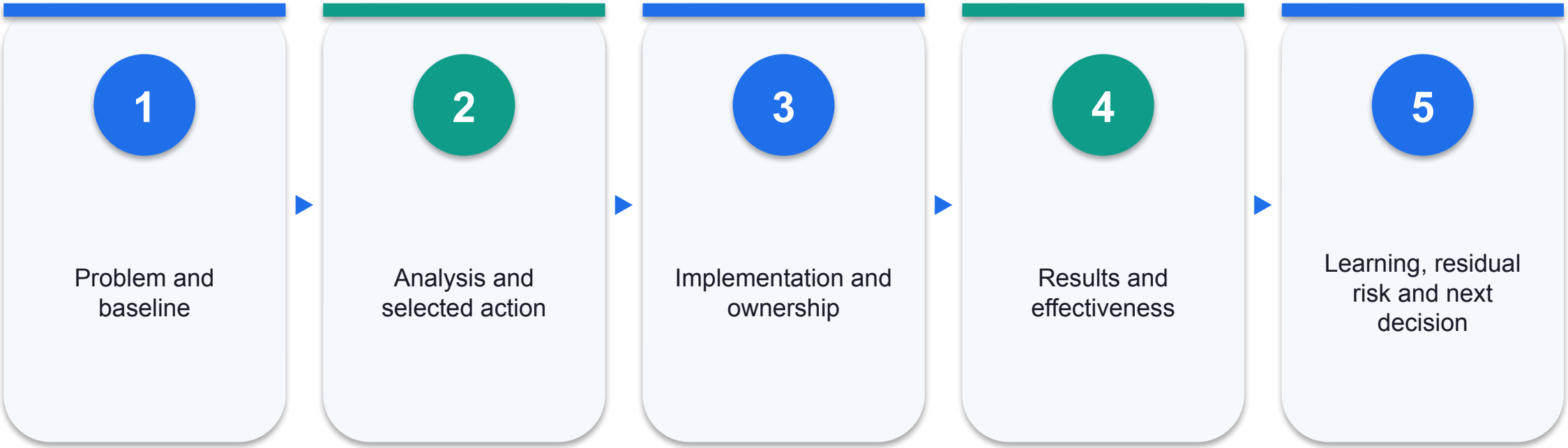
CHECK YOUR EVIDENCE

A reviewer can reconstruct the improvement story from evidence and clearly see ownership, status, approval, effectiveness and next review date.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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Improvement report storyline



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Strong improvement reporting

Build credibility

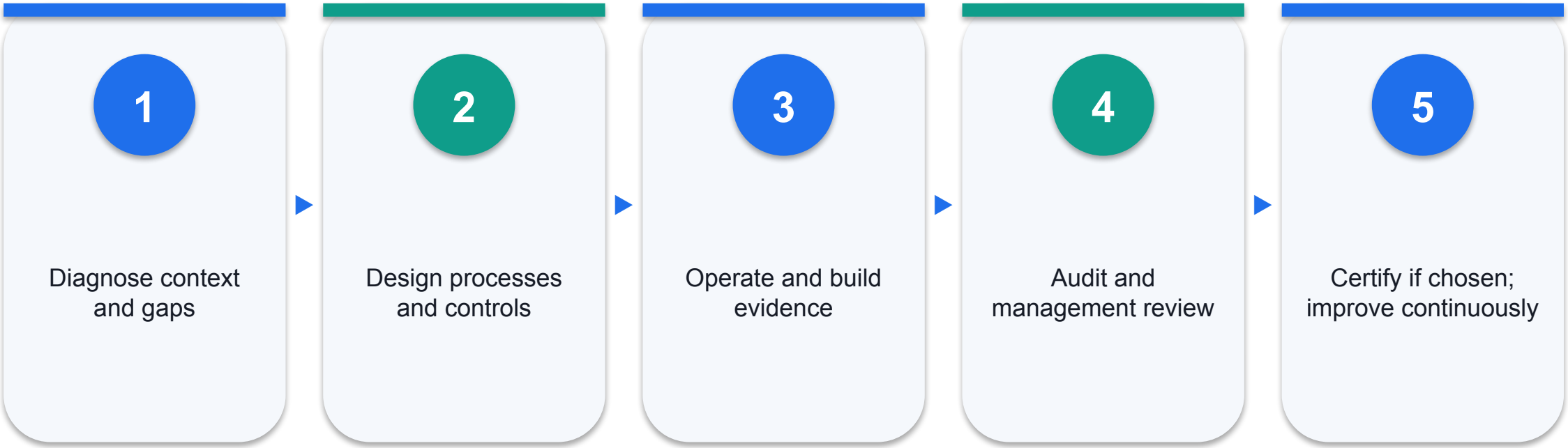
1. Traceable facts and data definitions
2. Clear owners, dates and approvals
3. Before-and-after comparison
4. Limitations and residual risks

Avoid

1. Success declared without baseline
2. Activity described without result
3. Late actions hidden
4. Lessons not converted into standard work

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Implementation roadmap



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COURSE SUMMARY

What you can now do

1. **Diagnose**
   Identify evidence-backed quality gaps.
2. **Map**
   Manage and improve end-to-end processes.
3. **Plan**
   Address risk and set measurable objectives.
4. **Control**
   Support and operate processes consistently.
5. **Evaluate**
   Use measures, audit and management review.
6. **Improve**
   Close the loop through CAPA and reporting.

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BEFORE YOU LEAVE

Course Feedback

1. Rate the content, activities and pace of the course.
2. Tell us what helped most and what we should improve.
3. Submit the feedback form shared by your trainer.
4. Apply one improvement from today at your workplace.

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THANK YOU

Build quality into the process.

Use evidence, ownership and continual improvement to make reliable performance repeatable.

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